

Status	Finished
Started	Thursday, 16 October 2025, 12:00 PM
Completed	Thursday, 16 October 2025, 12:30 PM
Duration	29 mins 20 secs
Marks	1.80/3.00
Grade	1.80 out of 3.00 (60.01%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

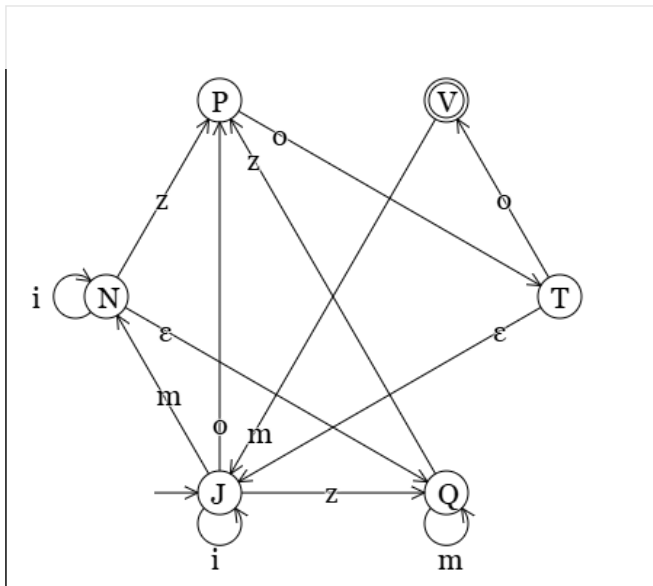
- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Correct

Mark 1.00 out of 1.00

Translate the following DFA to a right-regular grammar.



Write the answer as a 4-tuple, separating the rule per non-terminal. Use the names of the states for their respective nonterminals.

Your answers must be syntactically correct for marking.

Terminals = {"m","o","i","z"}

Your last answer was interpreted as follows:

{m, o, i, z}

Nonterminals = {"N","J","Q","T","V","P"}

Your last answer was interpreted as follows:

{N, J, Q, T, V, P}

Start symbol = "J"

Your last answer was interpreted as follows:

J

Rules for T = {["T", ["J"]], ["T", ["o", "V"]] }

Your last answer was interpreted as follows:

{[T, [J]], [T, [o, V]]}

Rules for Q = {["Q", ["z", "P"]], ["Q", ["m", "Q"]] }

Your last answer was interpreted as follows:

{[Q, [z, P]], [Q, [m, Q]]}

Rules for J = {["J", ["z", "Q"]], ["J", ["o", "P"]], ["J", ["m", "N"]], ["J", ["i", "J"]] }

Your last answer was interpreted as follows:

{[J, [z, Q]], [J, [o, P]], [J, [m, N]], [J, [i, J]]}

Rules for N = {["N", ["Q"]], ["N", ["z", "P"]], ["N", ["i", "N"]] }

Your last answer was interpreted as follows:

{[N, [Q]], [N, [z, P]], [N, [i, N]]}

Rules for P = {["P", ["o", "T"]] }

Your last answer was interpreted as follows:

{[P, [o, T]]}

Rules for V = {["V", ["m", "J"]], ["V", []] }

Your last answer was interpreted as follows:

{[V, [m, J]], [V, []]}

Question **2**

Incorrect

Mark 0.00 out of 1.00

Consider the following specification.

$$\{e < 10 \mid e, z := e', z' \mid z \geq 12\}$$

Which of the following implementations are correct with respect to the specification?

- ☐ $\{e < 8 \mid e := 10; z := e + 4 \mid z \geq 8\}$
- ☐ $\{e \leq 10 \mid e := 10; z := e + 4 \mid z \geq 8\}$
- ☐ $\{e < 10 \mid e := 10; z := e + 4 \mid z \geq 8\}$
- ☒ $\{e < 8 \mid e := 10; z := e + 4 \mid z \geq 12\}$
- ☒ $\{e < 10 \mid e := 10; z := e + 4 \mid z \geq 12\}$
- ☐ $\{e \leq 10 \mid e := 10; z := e + 4 \mid z \geq 12\}$
- ☐ $\{e \leq 10 \mid e := 10; z := e + 4 \mid z > 12\}$
- ☒ $\{e < 8 \mid e := 10; z := e + 4 \mid z > 12\}$
- ☒ $\{e < 10 \mid e := 10; z := e + 4 \mid z > 12\}$
- ☐ None of the above

Question **3**

Partially correct

Mark 0.80 out of 1.00

Compute the equivalence classes of the set S below with respect to the equivalence relation $\equiv_{11}$ (same remainder in division by 11).

$$S = \{-20, -18, -16, -12, -10, -7, -5, -3, -2, 3, 5, 6, 7, 10, 12, 13, 14, 16, 17, 19\}$$

The result must be given as a set of sets.

$$\{\{-20, 13\}, \{-18, 7\}, \{-16, -5, 6, 17\}, \{-12, 10\}, \{-10, 12\}, \{-3, 19\}, \{-2\}, \{3, 14\}, \{5, 16\}, \{7\}\}$$

Your last answer was interpreted as follows:

$$\{\{-20, 13\}, \{-18, 7\}, \{-16, -5, 6, 17\}, \{-12, 10\}, \{-10, 12\}, \{-3, 19\}, \{-2\}, \{3, 14\}, \{5, 16\}, \{7\}\}$$

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 302

Signing code:

Your last answer was interpreted as follows:

19605

Jump to...